

COURSE TITLE : **TIMBER STRUCTURES**
COURSE CODE : **6126**
COURSE CATEGORY : **E**
PERIODS PER WEEK : **4**
SEMESTER : **6**
PERIODS PER SEMESTER : **60**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1.	Structure Of Wood	6
	Structure Of Timber	7
	Test	1
2.	Design Structure Of Solid Timber	7
	Design Of Beam	8
	Test	1
3.	Design Of Column	4
	Design Of Joints	10
	Test	1
4.	Design Of Trusses	4
	Laminated Construction	8
	Proof Testing And Preservation	2
	Test	1
	TOTAL	60

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand Structure Of Solid Wood
	2	Understand Structure Of Timber
2	1	Understand Design Structure Of Solid Timber
	2	Understand Design Of Beam
3	1	Understand Design Of Columns
	2	Understand Design Of Joints
4	1	Understand Design Of Trusses
	2	Understand General Design Data

Specific outcome

MODULE I

1.1.0 Understand Structure of Solid Wood

- 1.1.1 Describe structure of wood
- 1.1.2 Differentiate softwood and hardwood.
- 1.1.3 Compare sapwood and heartwood.
- 1.1.4 Describe the factors affecting strength of wood.

1.2.0 Understand Structure of Timber

- 1.2.1 List the names of important softwoods and hardwoods.
- 1.2.2 Describe the properties of above timbers.
- 1.2.3 Describe the factors affecting selection of timber.

MODULE II

2.1.0 Understand Design Structure of Solid Timber

- 2.1.1 Describe design stresses for solid timber.
- 2.1.2 Describe extreme fiber stress in bending.
- 2.1.3 Describe basic working stresses.
- 2.1.4 Describe design stresses.
- 2.1.5 Describe condition of exposure, width of bearing, rate of growth.
- 2.1.6 Draw typical nailed trusses.

2.2.0 Understand Design of Beam

- 2.2.1 Describe solid beams.
- 2.2.3 Describe notched beams.
- 2.2.4 Describe built up beams.
- 2.2.5 Describe composite timber- concrete beams and slabs.

MODULE III

3.1.0 Understand Design of Columns

- 3.1.1 Describe solid rectangular columns.
- 3.1.2 Describe round columns.
- 3.1.3 Describe spaced columns.
- 3.1.4 Describe laminated columns.

3.2.0 Understand Design of Joints

- 3.2.1 Describe shrinkage effects with green timber and timer quality.
- 3.2.2 Describe nailed joints.
- 3.2.3 Describe splitting of timber.
- 3.2.4 Describe bolted joints.
- 3.2.5 Describe coach screws.
- 3.2.6 Describe glued joints.
- 3.2.7 Describe types of glued joints.
- 3.2.8 Describe factors affecting strength of glued joints.

MODULE IV

4.1.0 Understand Design of Trusses

- 4.1.1 Describe truss form and spacing.
- 4.1.2 Describe stiffness of timber trusses.
- 4.1.3 Describe depth to span ratio.
- 4.1.4 Calculation of deflection.
- 4.1.5 Describe basic working stresses for glued laminated timber.

4.2.0 Understand General Design Data

- 4.2.1 Describe end joints and edge joints.
- 4.2.2 Describe variation of grade within a member, mixed species.
- 4.2.3 Describe direction of sawing lamina.
- 4.2.4 Describe preservation of timber.
- 4.2.5 Describe methods of preservation.
- 4.2.6 Describe types of preservatives.
- 4.2.7 State the economy of preservative treatment.

CONTENT DETAILS

MODULE I

Structure of wood, comparison of softwood and hardwood, comparison of sapwood and heartwood, factors affecting strength of wood. – species, defects, moisture content, temperature, variability with in a species, duration of loading. Structure of timber, names of important Indian timbers,- density, shrinkage, tendency to warp, strength, durability, availability & cost properties of timber, selection of timber- species, strength properties, physical properties, durability, availability.

MODULE II

Design structure of solid timber, extreme fiber stress in bending, modulus of elasticity, compression parallel to grain, compression perpendicular to grain. Basic working stresses- grade and moisture content, design stresses, duration of loading, condition of exposure, angle of grain, width of bearing, rate of growth, diagram of typical nailed truss. Design of beams:- solid beam- fiber stress, form factor, longitudinal shear stress, stiffness, later stability, bearing stress (general idea only), solid beam, notched beams, built up beams, composite timber- concrete beams and slabs, design of laminated beams.

MODULE III

Design of columns, types of columns- rectangular columns, effective column length, round columns, spaced columns, design of laminated columns. Design of joints, types of connections, metal connectors- nail, screw and bolts, timber connectors, gluing. Shrinkage effects with green timber. Nailed joints, direction of loading, nails in end grain, moisture content, duration of loading. Multiple nail joints, nail penetration, splitting of timber, spacing of nails, allowable, withdrawal of loads. Bolted joints, multiple bolted joints, basic working loads parallel to grain and perpendicular to grain. Coach screws. Glued joints, factors affecting the strength of joints- timber quality, surface preparation, moisture content of wood and glue. Names of glues. Types of glued joints- bud joint, scarf joint, end joint, durability of glued joints.

MODULE IV

Truss form and spacing, arrangement of members, number of joints, spacing of joints, spacing of trusses, stiffness of timber trusses, depth to span ratio, calculation of deflection. Basic working stresses for glued laminated timber (table). Design stresses, duration of load, number of laminations, moisture content, condition of exposure, angle of grain. General design data- end joints- scarf joint, bud joint, edge joint. Variation of grade within a member, mixed species, thickness of lamination, glues, direction of sawing laminae. Preservation of timber structure- general idea, preservation methods, types of preservatives. Economy of preservative treatment.

REFERENCE

1. R.G.PERSON, Timber Engineering design hand book , The jacaranda press Australia
2. F.P.KOLLMAN ,Principles of wood science technology volume I&II,Springerverlag